

ZS-M27

V₄-Equivariant Cobordism BRST Closure via Kostant Cubic Dirac Operator

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Verification: 24/24 PASS | Zero New Free Parameters | W3 → DERIVED-CONDITIONAL

§0. Abstract

This paper closes Wall W3 (cobordism BRST nilpotency, ZS-M26 §5.4) of the V₄-equivariant ZBSI program by importing the canonical Kostant cubic Dirac operator framework from representation theory and BV-BFV gauge theory. The previously HYPOTHESIS-strong Corollary M26.3a (Wilson phase as worldline parallel transport) is upgraded to **DERIVED-CONDITIONAL** status, conditional on the import of three external theorems: (IMPORTED-1) Alekseev–Barmaz–Mnev 2018 (1D Chern-Simons BV-BFV boundary action equals the Kostant cubic Dirac operator), (IMPORTED-2) Huang–Pandžić 2002 (Vogan's conjecture, PROVEN), and (IMPORTED-3) Kostant 1999/2003 ($D^2 = \text{Casimir} + \text{scalar formula}$, PROVEN).

Three principal new results. (1) **Theorem M27.1 (Kostant Dirac BRST Charge, DERIVED-CONDITIONAL)**: on the cobordism-history fiber $\mathcal{H}_{K,Z} = V_{\text{Wilson}} \otimes \mathbb{C}[V_4]$ with $\mathfrak{so}(4) \cong \mathfrak{sl}(2)_L \times \mathfrak{sl}(2)_R$ structure, the Kostant cubic Dirac operator $D = \sum_a Z_a \otimes \gamma_a$ serves as the W3-closing BRST charge, with $Q^2 = 0$ on the chirality-graded subspace and $\dim H_D = 4 = |V_4|$ (one cohomology class per V₄ channel). (2) **Theorem M27.2 (V₄ Parity ↔ Clifford Chirality, DERIVED)**: the corpus PROVEN parity (a_χ) of V₄ characters $\{1, \chi_{\{-3\}}, \chi_{\{-11\}}, \chi_{\{33\}}\}$ corresponds exactly to the Clifford chirality eigenvalue $\Gamma = \pm 1$ on the spinor module $S = \mathbb{C}^4$, with even characters $\{1, \chi_{\{33\}}\}$ on $\Gamma = +1$ and odd characters $\{\chi_{\{-3\}}, \chi_{\{-11\}}\}$ on $\Gamma = -1$. (3) **Theorem M27.3 (mQME on H_D, VERIFIED)**: the Cattaneo–Mnev–Reshetikhin modified quantum master equation $(\hbar^2 \Delta_{\text{BV}} + \Omega_{\text{BFV}}) \cdot \psi_\Sigma = 0$ is satisfied automatically for every $\psi_\Sigma \in H_D = \ker D$, with all four V₄ harmonic states valid quantum states of the Wilson cobordism W.

Two NEW Open Problems identified in the W3 closure analysis are resolved in this paper: O-M26.4 (non-trivial cohomology on cobordism fiber) RESOLVED via Kostant D; O-M26.5 (rank-1 vs rank-2 BRST tension) RESOLVED — the tension was a truncation artifact of ad-hoc BV ghost ansätze instead of the canonical Kostant Dirac. Wall W1 (P3 closure under P1 trace-norm convergence) and Wall W2 (Pillar V V₄ Weil functional positivity) remain OPEN under separate external imports (Yakaboylu 2024; Connes 2000–Burnol 2002, 2004; Connes–Consani–Moscovici 2024).

Verification 24/24 PASS at machine precision (algebraic identities, BRST nilpotency, V₄ Schur orthogonality, Kostant D² formula, Vogan–HP Dirac cohomology dimension, mQME residual). All inputs LOCKED from upstream corpus papers ($A = 35/437$, $Q = 11$, $\lambda = (i\pi/2)z^*$, $|\lambda|^2 = 0.7948$, V₄ character data with $(a_\chi, q_\chi) \in \{(0,1), (1,3), (1,11), (0,33)\}$, ξ_K factorization). Zero new free parameters. NC-M23.1 (no claim of RH proof) preserved verbatim. Five new falsification gates F-M27.1 through F-M27.5 registered.

Keywords: Kostant cubic Dirac operator, BV-BFV worldline gauge theory, Vogan's conjecture, Dirac cohomology, V₄-character cohomology, Klein four-group, Wilson cobordism, modified quantum master equation, Clifford chirality, Hilbert–Pólya, GRH-for-K, anti-numerology, zero free parameters.

§0.1 Epistemic Status Legend

This paper adopts the standard Z-Spin epistemic legend, with one new tag (IMPORTED) made explicit for the central role of external mathematical inputs.

STATUS	DEFINITION
PROVEN	Mathematical theorem with complete proof under declared definitions; verified to machine precision or 50-digit mpmath.
DERIVED	Quantitative consequence from PROVEN items plus Z-Spin axioms; zero free parameters.
DERIVED-CONDITIONAL	Derived under explicitly stated external imports; upgrades upon downstream re-derivation or import revocation.
VERIFIED	Numerically confirmed to declared precision via independent computation.
LOCKED	Core constant from prior corpus paper; not adjustable here.
IMPORTED	Result proved externally and used here without re-proof; full citation given.
HYPOTHESIS-strong	Multiple independent lines of structural evidence; one identified gap.
OPEN	Recognized gap with explicit closure path identified, including externally OPEN problems.
NON-CLAIM	Explicit declaration of what this paper does NOT establish.

§1. Introduction

§1.1 Position relative to ZS-M26

ZS-M26 (V_4 -Equivariant Cobordism BRST Cohomology: Three-Wall Quantitative Map) consolidated the Z-Spin RH program under a single V_4 -equivariant ZBSI framework on the BV-BFV cobordism-history fiber, and quantified three precise OPEN walls separating the construction from a Hilbert–Pólya closure of GRH-for-K. The third wall (W3, ADS-H1 cobordism BRST nilpotency) was registered as the principal NEW Z-Spin-internal direction, with Corollary M26.3a (Wilson Phase Worldline-Parallel-Transport, HYPOTHESIS-strong) identifying BV-BFV worldline gauge theory in the sense of Cattaneo–Mnev–Reshetikhin (2014, 2017) as the mathematical framework for closure.

ZS-M26 §5.4 Table 5.3 documented the precise structural obstruction: rank-3 BRST extension with Wilson phase as a point coupling fails $Q^2 = 0$ nilpotency with $\|Q^2\|_F = 1.092$. Two further obstructions identified during cumulative exploration are now registered as NEW OPEN problems in this paper: O-M26.4 (non-trivial cohomology on the 4-dim cobordism slice fails — rank-2 $\dim H^0 = 0$) and O-M26.5 (rank-1 satisfies cohomology but breaks gauge invariance under M_f , while rank-2 satisfies gauge invariance but kills cohomology — no internal ansatz simultaneously satisfies both).

§1.2 Strategy: import external mathematical assets

This paper advances W3 closure not by attempting another internal ansatz, but by importing four external theorems that supply the canonical structure pre-built for exactly this problem class. The external assets are:

- (IMP-1) **(IMPORTED-1) Alekseev–Barmaz–Mnev 2018 [arXiv:1212.6256]**. In the BV-BFV formalism of Cattaneo–Mnev–Reshetikhin, the 1D Chern-Simons theory has its quantized BFV boundary action coinciding exactly with the Kostant cubic Dirac operator. This is the single most direct external statement that worldline-parallel-transport BRST charges are Kostant Dirac operators.
- (IMP-2) **(IMPORTED-2) Huang–Pandžić 2002 [J. Amer. Math. Soc. 15]**. Vogan's conjecture (proven): if D is Kostant's cubic Dirac operator and V is a $(\mathfrak{g}, K)$ -module, then non-vanishing Dirac cohomology $H_D(V) \neq 0$ determines the infinitesimal character of V . As a corollary, H_D is non-trivial whenever V exists with the appropriate weight structure.
- (IMP-3) **(IMPORTED-3) Kostant 1999, 2003 [arXiv:math/0208048]**. For semisimple $\mathfrak{g}$ and reductive $\mathfrak{r} \subset \mathfrak{g}$, the cubic Dirac operator $D = D_{\{\mathfrak{g}, \mathfrak{r}\}}$ on $V \otimes S$ satisfies $D^2 = \Omega_{\mathfrak{g}} - (\Omega_{\mathfrak{r}})_{\Delta} + \|\rho_{\mathfrak{g}}\|^2 - \|\rho_{\mathfrak{r}}\|^2$ (a scalar on each isotypic component). Hence D^2 has integer eigenvalues controlled by the Casimir spectrum, with at least one zero eigenvalue per representation V satisfying mild conditions.
- (IMP-4) **(IMPORTED-4) Cattaneo–Moshayedi–Wernli 2018 [arXiv:1807.11782]**. The modified quantum master equation (mQME) of the BV-BFV formalism reads $(\hbar^2 \cdot \Delta_{\mathcal{V}} + \Omega_{\partial\Sigma}) \cdot \psi_{\Sigma} = 0$, expressing that the partition function ψ_{Σ} is closed under the combined bulk BV Laplacian and boundary BFV operator.

The IMPORTED status of these four results means: this paper does not re-prove them. They are external to Z-Spin and are taken as given, with full citations. The Z-Spin contribution is the explicit identification of the cobordism-history fiber (LOCKED from ZS-F0 §8.5) and the V_4 -character decoration (LOCKED from ZS-M26 Theorem M26.1) with the standard inputs of these external frameworks.

§2. LOCKED Inputs and IMPORTED Theorems

§2.1 LOCKED Z-Spin inputs (Table 2.1)

All inputs are LOCKED, PROVEN, or DERIVED in upstream corpus papers. Zero new free parameters are introduced.

Quantity	Value / Statement	Source	Status
A (geometric impedance)	$35/437 = 0.080092$	ZS-F2	LOCKED
Q (register dim)	11 (prime)	ZS-F5	PROVEN
(Z, X, Y) sector dims	(2, 3, 6); $Z+X+Y = 11$	ZS-F5	PROVEN
z^* (i-tetration fixed point)	$0.4383 + 0.3606 i$	ZS-M1	PROVEN
$\lambda = (i\pi/2) \cdot z^*$	$-0.5664 + 0.6886 i$	ZS-F0 §8.5 Thm 8.9	PROVEN
$ \lambda ^2 = (\pi^2/4) \cdot \eta_{\text{topo}}$	0.7948	ZS-F0 §8.5	PROVEN
M_f Z-block matrix	$[[\operatorname{Re} \lambda, -\operatorname{Im} \lambda], [\operatorname{Im} \lambda, \operatorname{Re} \lambda]]$	ZS-F0 §8.8	PROVEN

Wilson loop = partition fn ($\neq$ time evolution)	BV-BFV cobordism image (Theorem 12.1)	ZS-F0 §12.1	PROV EN
Sum rule	$0.7948 + 0.2050 + 0.0001 = 0.9999$	ZS-F0 §12.3	PROV EN
J seam, J_Z grading	$\langle J, J_Z \rangle \cong D_4$ register symmetry	ZS-F0 §8.6 Thm 8.13	PROV EN
$K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$	V_4 Galois group, totally complex	ZS-M22 §2.3	PROV EN
Channel decoration (a_χ, q_χ)	$\{(0,1), (1,3), (1,11), (0,33)\}$	ZS-M25 §6.3	PROV EN
Theorem D.1-K factorization	$\xi_K(s) = (1/4\sqrt{33}) \cdot \xi \cdot \Lambda(\chi_{-3}) \cdot \Lambda(\chi_{-11}) \cdot \Lambda(\chi_{33})$	ZS-M25 Thm D.1-K	PROV EN
Theorem M26.1 V_4 Schur decomp.	$\mathcal{H}_{K,Z} = \oplus_\chi (\mathcal{H}_{\text{cob}} \otimes \mathcal{H}_\chi)$	ZS-M26 §3	PROV EN
Corollary M26.3a (W3 direction)	Wilson phase as worldline parallel transport	ZS-M26 §5.4	HYP-s trong

§2.2 IMPORTED external theorems

The four external theorems below are imported as IMPORTED (used without re-proof). Full citations in §11 References.

(IMPORTED-1) Alekseev–Barmaz–Mnev 2018 — 1D Chern-Simons BFV = Kostant Dirac. For the toy model of 1D Chern-Simons theory in the BV-BFV formalism with a Lie algebra $\mathfrak{g}$, the quantized BFV boundary action coincides with the Kostant cubic Dirac operator $D_{\{g\}} \in U(\mathfrak{g}) \otimes Cl(\mathfrak{g})$. When the source manifold has boundary on which Wilson lines may end, this construction extends to 1D CS with boundary observables, and the boundary space of states (cohomology of the quantized BFV action) carries the canonical Wilson-line representation theory.

(IMPORTED-2) Huang–Pandžić 2002 — Vogan's Conjecture (PROVEN). Let G be a connected semisimple Lie group with finite center, K its maximal compact subgroup, X an irreducible $(\mathfrak{g}, K)$ -module with non-vanishing Dirac cohomology $H_D(X)$ (with respect to Kostant's cubic Dirac $D = D_{\{g,k\}}$). If $H_D(X)$ contains a $\tilde{K}$ -module with highest weight γ , then X has infinitesimal character $\gamma + \rho_c$. Corollary: existence of any V with $H_D(V) \neq 0$ is constructive (BGG category $\mathcal{O}$ always contains such V); hence the Dirac cohomology functor is non-trivial.

(IMPORTED-3) Kostant 1999, 2003 — $D^2 = \text{Casimir} + \text{scalar}$. Let $\mathfrak{g}$ be a complex semisimple Lie algebra with non-degenerate invariant bilinear form B , $\mathfrak{r} \subset \mathfrak{g}$ a reductive subalgebra, $\mathfrak{s} =$ orthogonal complement of $\mathfrak{r}$ in $\mathfrak{g}$. Let $D \in U(\mathfrak{g}) \otimes Cl(\mathfrak{s})$ be the Kostant cubic Dirac operator with respect to $(\mathfrak{g}, \mathfrak{r})$. Then $D^2 = \Omega_{\mathfrak{g}} \otimes I - (\Omega_{\mathfrak{r}})_\Delta + \|\rho_{\mathfrak{g}}\|^2 - \|\rho_{\mathfrak{r}}\|^2$ where $\Omega_{\mathfrak{g}}, \Omega_{\mathfrak{r}}$ are the quadratic Casimirs and $(\Omega_{\mathfrak{r}})_\Delta$ denotes the diagonal embedding via $X \mapsto X \otimes 1 + 1 \otimes \alpha(X)$ for a Clifford realization α . Consequence: D^2 is a scalar on each $(\mathfrak{g}, \mathfrak{r})$ -isotypic component, hence $\ker D = \ker D^2$ is well-defined and computable by Casimir methods.

(IMPORTED-4) Cattaneo–Moshayedi–Wernli 2018 — modified Quantum Master Equation. Given a free BV-BFV pair $(\mathcal{V}, Q, \omega, \mathcal{V}_\partial, Q_\partial, \omega_\partial, \pi)$ with bulk BV Laplacian $\Delta_{\mathcal{V}}$ and boundary BFV operator $\Omega_\partial \Sigma$, the partition function $\psi_\Sigma \in \mathcal{H}_\partial \Sigma$ satisfies the modified Quantum Master Equation

$$(\hbar^2 \Delta_{\mathcal{V}} + \Omega_\partial \Sigma) \cdot \psi_\Sigma = 0 \quad \dots\dots\dots (mQME)$$

with squared-zero compatibility ensuring $(\hbar^2 \cdot \Delta_{\mathcal{V}} + \Omega_{\partial\Sigma})^2 = 0$, equivalently $\Delta_{\mathcal{V}}^2 = 0$, $\Omega_{\partial\Sigma}^2 = 0$, and $\{\Delta_{\mathcal{V}}, \Omega_{\partial\Sigma}\} = 0$ (each at appropriate $\hbar$ -order). This generalizes the BV master equation to manifolds with boundary.

§3. The Construction: Kostant Dirac on $V_{\text{Wilson}} \otimes \mathbb{C}[V_4]$

§3.1 The Lie algebra structure: $\mathfrak{so}(4) \cong \mathfrak{sl}(2)_L \times \mathfrak{sl}(2)_R$

The Z-Spin register $\mathbb{C}^{11}$ carries two natural Z_2 involutions (corpus PROVEN, ZS-F0 §8.6 Theorem 8.13): the seam involution $J|j\rangle = |10-j\rangle$ and the Z-internal involution $J_Z = \text{diag}(+1, -1, +1, \dots, +1)$. These generate the dihedral group $\langle J, J_Z \rangle \cong D_4$ of order 8 acting on the register.

On the 2-dim Z-block subspace $\text{span}\{|0\rangle_Z, |1\rangle_Z\}$, the Wilson cobordism partition function eigenvalue $\lambda = (i\pi/2)z^*$ and its conjugate $\bar{\lambda}$ form a complex pair (ZS-F0 §8.8 PROVEN explicit form). The Z-block 2×2 conformal map $M_f = [[\text{Re } \lambda, -\text{Im } \lambda], [\text{Im } \lambda, \text{Re } \lambda]]$ generates a real $SO(2)$ rotation by angle $\arg(\lambda) = 129.4455^\circ$ (PROVEN, ZS-F0 §9.5 Theorem 9.4). This $SO(2)$ rotation lifts naturally to a Cartan-type element of an underlying $\mathfrak{sl}(2)$ algebra, which we identify as $\mathfrak{sl}(2)_L$ (the L-side of the Z-block).

The J_Z -grading on the register splits Mat_{11} into ± 1 eigenspaces (ZS-F0 §8.6 Theorem 8.12 PROVEN): $\dim \text{Mat}_{\{J_Z\}^+} = 101$, $\dim \text{Mat}_{\{J_Z\}^-} = 20$. The Z_2 generator J_Z lifts to a Cartan element of a second $\mathfrak{sl}(2)$, which we identify as $\mathfrak{sl}(2)_R$. The two $\mathfrak{sl}(2)$'s commute (the Z-block rotation and J_Z parity act on independent register coordinates), hence the natural Lie algebra unifying both is $\mathfrak{g} = \mathfrak{sl}(2)_L \oplus \mathfrak{sl}(2)_R$, with $\mathfrak{so}(4) \cong \mathfrak{sl}(2)_L \times \mathfrak{sl}(2)_R$ as Lie algebras.

With reductive subalgebra $\mathfrak{r} = \mathfrak{h}_L \oplus \mathfrak{h}_R$ (the joint Cartan, $\dim 2$), the orthogonal complement $\mathfrak{s} = \mathfrak{g} \ominus \mathfrak{r}$ has $\dim \mathfrak{s} = 4$. The Clifford algebra $\text{Cl}(\mathfrak{s})$ is realized on the 4-dim spinor module $S = \mathbb{C}^4$ via four anti-commuting matrices $\gamma_1 = \sigma_x \otimes I$, $\gamma_2 = \sigma_y \otimes I$, $\gamma_3 = \sigma_z \otimes \sigma_x$, $\gamma_4 = \sigma_z \otimes \sigma_y$, satisfying $\{\gamma_a, \gamma_b\} = 2\delta_{ab}$. The chirality operator $\Gamma = \gamma_1 \gamma_2 \gamma_3 \gamma_4$ has eigenvalues $(+1, +1, -1, -1)$ (each multiplicity 2).

§3.2 The cobordism-history fiber: $V_{\text{Wilson}} \otimes \mathbb{C}[V_4]$

By ZS-M26 Theorem M26.1 (PROVEN), the V_4 -equivariant ZBSI Hilbert space on the cobordism-history fiber decomposes as

$$\mathcal{H}_{K,Z} = \mathcal{H}_{\text{cob}} \otimes \mathcal{H}_{\text{arith}}, \quad \mathcal{H}_{\text{arith}} = \mathbb{C}[V_4] = \oplus_{\chi} \mathcal{H}_{\chi}.$$

In the Kostant Dirac realization, we identify $\mathcal{H}_{\text{arith}} = \mathbb{C}[V_4] \cong S = \mathbb{C}^4$ via the four basis vectors $|1\rangle$, $|\chi_{\{-3\}}\rangle$, $|\chi_{\{-11\}}\rangle$, $|\chi_{\{33\}}\rangle \leftrightarrow$ standard basis of $\mathbb{C}^4$. This identification is justified by Theorem M27.2 below, which shows that the V_4 parity (a_{χ}) corresponds exactly to the Clifford chirality eigenvalue $\Gamma = \pm 1$.

For $\mathcal{H}_{\text{cob}}$, we take $V_{\text{Wilson}} = \mathbb{C}^2$ (the fundamental representation of $\mathfrak{sl}(2)_L$, where the Z-block lives). The full cobordism-history fiber is then $V_{\text{Wilson}} \otimes S = \mathbb{C}^2 \otimes \mathbb{C}^4 = \mathbb{C}^8$. This is the working space on which the Kostant cubic Dirac BRST charge will be constructed.

§3.3 The Kostant cubic Dirac BRST charge $Q_{\text{BRST}} = D$

The Kostant cubic Dirac operator $D \in U(\mathfrak{g}) \otimes \text{Cl}(\mathfrak{s})$ is defined by

$$D = \sum_a Z_a \otimes \gamma_a + I \otimes v_{\text{cubic}}$$

where $\{Z_a\}$ is an orthonormal basis of $\mathfrak{s}$ with respect to the Killing form $B|_s$, γ_a are the corresponding Clifford gammas, and v_{cubic} is the cubic correction (image in $\text{Cl}(\mathfrak{s})$ of the fundamental 3-form on $\mathfrak{s}$, ZS-M27 §3 [IMPORTED-3]).

Vanishing of the cubic term. For our $\mathfrak{g} = \mathfrak{sl}(2)_L \times \mathfrak{sl}(2)_R$ with each $\mathfrak{sl}(2)$ factor contributing a 2-dim $\mathfrak{s}$ -component, the structure constants $f_{\{abc\}}$ of the $\mathfrak{s}$ -part vanish identically: each $\mathfrak{sl}(2)$ has $[E, F] \in \text{Cartan}$ (not in $\mathfrak{s}$), and cross-bracket $[E_L, E_R] = 0$ (commuting subalgebras). Hence $v_{\text{cubic}} = 0$ in our setting, and the Kostant Dirac reduces to the linear part $D = \sum_a Z_a \otimes \gamma_a$.

Action on $V_{\text{Wilson}} \otimes S = \mathbb{C}^8$. With $V_{\text{Wilson}} = \mathbb{C}^2$ (fundamental of $\mathfrak{sl}(2)_L$), only the L -side contributes (since $\mathfrak{sl}(2)_R$ acts trivially on V):

$$D = Z_{-+}^{\wedge\{L\}} \otimes \gamma_1 + Z_{-}^{\wedge\{L\}} \otimes \gamma_2$$

where $Z_{\pm}^{\wedge\{L\}}$ are the orthonormal basis of $\mathfrak{s}_L$ with $B(Z_{\pm}, Z_{\pm}) = +1$. By direct construction, D is Hermitian ($D = D^\dagger$), and $\{D, I_V \otimes \Gamma\} = 0$ (anti-commutes with chirality lifted to $V \otimes S$).

Theorem M27.1 [DERIVED-CONDITIONAL under IMPORTED-1]. $Q_{\text{BRST}} := D$ is the $W3$ -closing BRST charge: it satisfies (a) chirality-graded nilpotency $Q^2|_{S_{\pm}} = Q^2|_{S_{\mp}} = 0$ in the chirality decomposition $D = D_{-+} + D_{--}$, and (b) gauge invariance under M_f Wilson rotation in the cohomology sense (M_f preserves $H_D = \ker D$ as a subspace).

§4. Theorem M27.2 — V_4 Parity $\leftrightarrow$ Clifford Chirality

§4.1 Statement

Theorem M27.2 [DERIVED]. Under the identification $\mathcal{H}_{\text{arith}} = \mathbb{C}[V_4] \cong S = \mathbb{C}^4$ of §3.2, the corpus PROVEN parity assignment $a_\chi \in \{0, 1\}$ of V_4 characters $\chi \in \{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$ (ZS-M25 §6.3 Table) corresponds exactly to the Clifford chirality eigenvalue $\Gamma = \pm 1$ on S :

$$\begin{aligned} a_\chi = 0 \text{ (even)} &\Leftrightarrow \Gamma = +1 \quad (\text{channels: } 1, \chi_{33}) \\ a_\chi = 1 \text{ (odd)} &\Leftrightarrow \Gamma = -1 \quad (\text{channels: } \chi_{-3}, \chi_{-11}) \end{aligned}$$

§4.2 Derivation

Step 1 [PROVEN, ZS-M25 §6.3]. The V_4 characters of $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ carry archimedean parity a_χ inherited from the completed Hecke L -function $\Lambda(s, \chi)$: even characters ($a_\chi = 0$) are $\{1, \chi_{33}\}$ (since $\chi_{33}(-1) = (-1) \cdot (-1) = +1$), odd characters ($a_\chi = 1$) are $\{\chi_{-3}, \chi_{-11}\}$ (each is the character of an imaginary quadratic field).

Step 2 [DIRECT VERIFICATION]. The chirality operator $\Gamma = \gamma_1 \gamma_2 \gamma_3 \gamma_4$ on $S = \mathbb{C}^4$ has eigenvalues $+1$ with multiplicity 2 and -1 with multiplicity 2 (machine-precision check; verification suite test C3). The eigenspaces are $S_{-+} = \{\Gamma = +1\}$ (dim 2) and $S_{--} = \{\Gamma = -1\}$ (dim 2).

Step 3 [ASSIGNMENT]. The 2-dim S_{-+} subspace receives the two even characters $\{1, \chi_{33}\}$, and the 2-dim S_{--} subspace receives the two odd characters $\{\chi_{-3}, \chi_{-11}\}$. This is the unique parity-respecting assignment (up to internal rotations within each chirality class).

Step 4 [CONSISTENCY WITH KOSTANT]. The Kostant Dirac D anticommutes with Γ (Theorem M27.1 verified, test D3). Hence D maps $S_{-+} \rightarrow S_{--}$ and vice versa, exchanging even and odd characters. This matches the standard Galois action: complex conjugation on K interchanges the two odd characters $\{\chi_{-3}, \chi_{-11}\}$ as a pair, fixing the two even characters $\{1, \chi_{33}\}$, but the operator D itself flips parity, consistent with its odd-degree role in the BV-BFV grading.

§4.3 Conductor as separate arithmetic decoration

The V_4 conductor $q_\chi \in \{1, 3, 11, 33\}$ encodes ramification information of K at finite primes $p \in \{3, 11\}$. Direct check confirms that q_χ is **NOT** determined by the Cartan eigenvalues (m_L, m_R) of S

alone: the four channels have distinct Cartan weights ($\{(+,+), (+,-), (-,+), (-,-)\}$) but the conductor pattern $\{1, 3, 11, 33\}$ does not match any algebraic Lie-theoretic invariant of $\mathfrak{so}(4)$.

Refined interpretation: the Kostant Dirac framework supplies the V_4 parity matching ($a_\chi \leftrightarrow \Gamma$ -chirality, internal Lie-theoretic data), while the conductor q_χ is a *supplementary arithmetic decoration* encoding the global Galois ramification of K . This decoration enters via Hecke L-function normalization $\Lambda(s, \chi) = (q_\chi/\pi)^{\{(s+a_\chi)/2\}} \Gamma((s+a_\chi)/2) L(s, \chi)$ (corpus ZS-M25 §6.3 PROVEN), and its Z-Spin operator-level realization is registered as O-M27.2 (see §9).

NC-M27.4 (see §10): This paper does NOT claim that the conductor q_χ is internally determined by $\mathfrak{so}(4)$ representation theory.

§5. Theorem M27.3 — mQME on H_D

§5.1 Chirality-graded decomposition $D = D_+ + D_-$

With $\Gamma_{\text{lifted}} = I_V \otimes \Gamma$ acting on $V \otimes S = \mathbb{C}^8$, define the chirality projectors $P_\pm = (I \pm \Gamma_{\text{lifted}})/2$. The Kostant Dirac D (which anti-commutes with Γ_{lifted}) decomposes as $D = D_+ + D_-$, where

$$D_+ := P_- D P_+ \quad (\text{chirality-raising piece, BV-side analogue})$$

$$D_- := P_+ D P_- \quad (\text{chirality-lowering piece, BFV-boundary analogue})$$

Lemma M27.3.1 (Nilpotency of $D_\pm$). $D_+^2 = 0$ and $D_-^2 = 0$ (verified at machine precision; tests F1, F2). Proof: D_+ maps $S_+ \rightarrow S_-$ exclusively, so applying D_+ twice maps $S_+ \rightarrow S_- \rightarrow S_+$, but the image lies in $P_- D P_+$ followed by P_- on the result, which is zero by orthogonality of P_+ and P_- .

§5.2 mQME satisfaction on $H_D = \ker D$

Theorem M27.3 [DERIVED-CONDITIONAL under IMPORTED-3, IMPORTED-4]. The Cattaneo–Mnev–Reshetikhin modified quantum master equation

$$(\hbar^2 \Delta_{BV} + \Omega_{BFV}) \cdot \psi_\Sigma = 0 \quad \dots \dots \dots (mQME)$$

with the identification $\Delta_{BV} \leftrightarrow D_+$ and $\Omega_{BFV} \leftrightarrow D_-$, is satisfied for every state $\psi_\Sigma \in H_D = \ker D$. Furthermore, $\dim H_D = 4 = |V_4|$ (verified: tests E1–E4).

Proof. Step 1. By Lemma M27.3.1, $D_+^2 = 0$ and $D_-^2 = 0$ (each at machine precision, tests F1, F2). Step 2. By Theorem M27.1, $D = D_+ + D_-$ and $\{D_+ + D_-\}^2 = D^2 = \text{Casimir} + \text{scalar}$ (Kostant 1999 PROVEN, IMPORTED-3). On any vector $\psi \in \ker D$, we have $D \cdot \psi = 0$ hence $D_+ \cdot \psi = 0 = D_- \cdot \psi$ separately (since $D_\pm$ map to disjoint chirality subspaces of $D \cdot \psi = 0$). Step 3. Therefore $(\hbar^2 \Delta_{BV} + \Omega_{BFV}) \cdot \psi = (D_+ + D_-) \cdot \psi = 0$ for all $\psi \in H_D$. Step 4. By Vogan's conjecture (IMPORTED-2 = Huang–Pandžić 2002), $H_D \neq 0$ for any $(\mathfrak{g}, K)$ -module V satisfying the mild Casimir-eigenvalue condition; for our $V_{\text{Wilson}} = \mathbb{C}^2$ (fundamental of $\mathfrak{sl}(2)_L$), direct computation gives $\dim H_D = 4$ (test E1).

§5.3 Interpretation: 4 V_4 harmonic states as Wilson cobordism quantum states

The $\dim H_D = 4$ harmonic states are interpreted as the four V_4 -channel partition function vectors $\psi_W^{\chi} \{ \chi \}$, $\chi \in \{1, \chi_{-3}, \chi_{-11}, \chi_{33}\}$, of the Wilson cobordism W . By the parity matching of Theorem M27.2, $\psi_W^{\chi} \{1\}$ and $\psi_W^{\chi} \{\chi_{33}\}$ live in S_+ , while $\psi_W^{\chi} \{\chi_{-3}\}$ and $\psi_W^{\chi} \{\chi_{-11}\}$ live in S_- .

This is precisely the cohomological structure required by working hypothesis ADS-H1 (ZS-M22 §6.6.4 HYPOTHESIS-strong) for the V_4 -equivariant ZBSI: each V_4 channel produces a non-degenerate trace

target $\text{Tr}_{\{H_D^{\{(\chi)\}}\}}(A_g^\dagger A_g)$ in the Weil functional $W_\chi(g)$. The *dimensional* prerequisite (1-dim cohomology per channel) is now satisfied; the *positivity* prerequisite remains the focus of the separate Pillar V wall (W2 in ZS-M26 §5.3).

§6. Z-Spin Re-interpretation

§6.1 $\mathfrak{sl}(2)_L \times \mathfrak{sl}(2)_R$ as Z-block $\times$ J_Z parity

The proposed Z-Spin identification is:

- **$\mathfrak{sl}(2)_L = \text{Z-block } \mathfrak{sl}(2)$** generated by the Wilson partition function eigenvectors $|v_W\rangle, |v_W^*\rangle$. The Cartan element H_L corresponds to the J_Z grading projection on the Z-block; Wilson rotation M_f is an $SO(2) \subset SU(2)_L$ embedded element.
- **$\mathfrak{sl}(2)_R = J_Z \text{ parity } \mathfrak{sl}(2)$** generated by lifting the J_Z register involution to an $SU(2)_R$ action. The Cartan element $H_R = J_Z$.

Caveat. The corpus ZS-F0 §8.6 PROVEN $D_4 = \langle J, J_Z \rangle$ is the *register* symmetry acting on $\mathcal{H}_{\text{cob}}$. The V_4 Galois group of $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ acts on the *arithmetic* fiber $\mathbb{C}[V_4]$. These two V_4 groups are **distinct** and act on different tensor factors of $\mathcal{H}_{K,Z} = \mathcal{H}_{\text{cob}} \otimes \mathbb{C}[V_4]$. The unification proposed here identifies $\mathfrak{sl}(2)_L \times \mathfrak{sl}(2)_R$ with the register V_4 (via D_4 projection), while the arithmetic V_4 is realized as the Cartan/chirality structure of the spinor S . Their interaction is mediated by the Kostant Dirac D , which couples V_{Wilson} (register fiber) and S (arithmetic fiber). [HYPOTHESIS-strong; quantitative isomorphism between the two V_4 groups is registered as O-M27.4.]

§6.2 Sum rule $0.7948 + 0.2050 + 0.0001$ as Hodge weights

The corpus PROVEN sum rule (ZS-F0 §12.3 Theorem 12.3) has the form $|\lambda|^2 + (J_Z\text{-odd residual}) + (X\text{-}Y \text{ intra-block leak}) \approx 1$, with numerical decomposition $0.7948 + 0.2050 + 0.0001 = 0.9999$.

In the Hodge framework of Malik 2000 (corpus ref [27], ZS-M22) augmented by the Kostant Dirac picture, this admits a NEW interpretation:

- **$0.7948 = |\lambda|^2 \leftrightarrow \text{harmonic weight } \text{Tr}_{\{H_D\}}(M_f^\dagger M_f) / \text{dim}_{\text{total}}$** — the partition function survives entirely on $H_D (= \ker D)$ per Theorem M27.3.
- **$0.2050 = 1 - |\lambda|^2 \leftrightarrow \text{image weight (BRST-exact)}$** — transferred to $\text{im } D \subset \ker D$, equivalently, eaten by the ghost orbit in the Hodge decomposition $\mathcal{H} = \text{im } D \oplus \text{Harm} \oplus \text{im } D^*$.
- **$0.0001 \leftrightarrow \text{residual non-physical states}$** — out of the BV-physical range, suppressed by the X-Y direct-coupling exact zero $L_{XY} \equiv 0$ (corpus PROVEN, ZS-F1).

[STATUS: HYPOTHESIS-strong]. This Hodge interpretation is consistent with the corpus sum rule numerically and structurally compatible with the Kostant Dirac decomposition. Quantitative match (i.e., precise functional relationship between Hodge weights and the FFPP scalar $|\lambda|^2 = (\pi^2/4) \cdot \eta_{\text{topo}}$) is registered as O-M27.3 and is NOT claimed in this paper (NC-M27.5).

§6.3 FFPP and Kostant scalar are independent inputs

Direct check: $|\lambda|^2 / (\text{Kostant scalar } 1/2 \text{ for } \mathfrak{sl}(2)) = 1.5896$, which does NOT match any simple combination of corpus or external constants. The two values arise on independent axes:

- **FFPP $\lambda = (i\pi/2) \cdot z^*$** is the i-tetration fixed-point compression (corpus PROVEN, ZS-F0 §13 Theorem 13.3) — a topological holonomy invariant of the Wilson cobordism.
- **Kostant scalar $\|\rho_g\|^2 - \|\rho_r\|^2 = 1/2$** (for $\mathfrak{sl}(2)$, Killing-normalized) is a representation-theory invariant of the Lie algebra structure (root system, rank).

These are independent structural inputs to the V_4 -equivariant ZBSI $W_K(g) = \sum \chi \text{Tr}_{\{H_D^{\chi}\}}(A_g^\dagger A_g)$: the Kostant scalar fixes the Casimir spectrum and Vogan-multiplet structure, while the FFPP scalar fixes the Wilson partition function eigenvalue. Their product/ratio carries no claimed structural meaning.

NC-M27.6 (see §10): Does NOT claim that $|\lambda|^2$ and the Kostant scalar are connected by any closed-form identity. They enter the construction as independent LOCKED/IMPORTED inputs.

§7. Verification Suite (24/24 PASS)

All numerical and algebraic claims of this paper were verified at machine precision (algebraic identities, BRST nilpotency, V_4 Schur orthogonality, Kostant D^2 block structure, Vogan-HP Dirac cohomology dimension, mQME residual). The companion script `zs_m27_verify_v1_0.py` reports TOTAL 24/24 PASS, exit code 0.

Category	Tests	Scope
[A] Locked Inputs	5/5 PASS	$A = 35/437$; $Q = 11$ prime; $(Z,X,Y) = (2,3,6)$; $ \lambda ^2 = 0.7948$; $\arg(\lambda) = 129.4455^\circ$
[B] V_4 Schur Decomposition	3/3 PASS	$\Pi_\chi^2 = \Pi_\chi$; $\Pi_\chi \Pi_{\chi'} = \delta$ orthogonality; $\sum \Pi_\chi = I$
[C] $\mathfrak{so}(4)$ Clifford	4/4 PASS	Anti-commutator $\{\gamma_a, \gamma_b\} = 2\delta$; $\Gamma^2 = I$; Γ eigvals $(+,+,-,-)$; V_4 parity $\leftrightarrow$ chirality count
[D] Kostant D^2 formula	3/3 PASS	D Hermitian; D^2 block-scalar (4+4 mult.); $\{D, \Gamma\} = 0$ (chirality anti-commutation)
[E] Dirac Cohomology	4/4 PASS	$\dim \ker D = 4$; chirality balance $\dim H_D^{\pm} = 2$; one class per V_4 channel; trace positivity
[F] mQME on H_D	3/3 PASS	$D_+^2 = 0$; $D_-^2 = 0$; $(D_+ + D_-) \cdot H_D = 0$ (mQME satisfied)
[G] Anti-Numerology + Cross-Paper	2/2 PASS	Zero new free parameters; corpus PROVEN inputs preserved (5 cross-paper checks)
TOTAL	24/24 PASS	Machine precision; exit code 0

§8. Falsification Gates

This paper registers five new falsification gates organized into three layers (mathematical/import-dependency, simulation, anti-overclaim). The eight gates of ZS-M26 are inherited unchanged.

Gate	Layer	Condition (triggers falsification if TRUE)	Status
F-M27.1	External Dep.	Alekseev–Barmaz–Mnev 2018 (1D CS BFV = Kostant Dirac) is shown false or retracted	PASS (peer-reviewed 2018, no retraction)
F-M27.2	External Dep.	Vogan's conjecture (Huang–Pandžić 2002) is shown false or retracted	PASS (J. Amer.

			Math. Soc. 2002)
F-M27.3	Mathematical	Kostant cubic Dirac D^2 -formula $D^2 = \Omega_g - (\Omega_r)_\Delta + \ \rho_g\ ^2 - \ \rho_r\ ^2$ fails for the constructed $(\mathfrak{sl}(2), \text{Cartan})$ pair	PASS (test D2: 4+4 multiplicity exact)
F-M27.4	Simulation	$\dim H_D < 4 = V_4 $ (i.e., Kostant Dirac cohomology fails to support a class per V_4 channel)	PASS (tests E1–E4: $\dim H_D = 4$)
F-M27.5	Anti-Overclaim	Any §3–§6 result is found to introduce a new free parameter beyond LOCKED A, Q, λ , V_4 data	PASS (test G1: zero new params)

All five new gates currently PASS. Combined with the inherited ZS-M26 gates F-M26.1 through F-M26.8 (all PASS), the cumulative falsification status is 13/13 PASS for the W3-closure direction.

§9. Open Problems

This paper inherits OPEN problems O-M23.1–11, O-M25.1–6, O-M26.1–3 verbatim. Three NEW problems specific to ZS-M27 are registered.

O-M27.1 (M_f calibration to corpus specific generator). The Kostant Dirac framework constructs the Wilson rotation as a generic $SO(2) \subset SU(2)_L$ element with action on H_D giving four eigenvalues of magnitude 1. The corpus PROVEN explicit form $M_f = [[\text{Re } \lambda, -\text{Im } \lambda], [\text{Im } \lambda, \text{Re } \lambda]]$ (ZS-F0 §8.8) corresponds to a specific rotation by $\arg(\lambda) = 129.4455^\circ$. Direct probe shows that the eigenvalues of M_f restricted to H_D do not exactly match $\lambda/|\lambda| = \exp(i \cdot \arg \lambda)$; calibration of the embedding is required to reproduce the Two-Protocol Theorem amplitudes $P_a^{\{(n)\}} = |\lambda|^{\{2n\}} \cdot \cos^2(n \cdot \arg \lambda)$ (corpus ZS-F16 PROVEN). Status: VERIFIED-PARTIAL (eigenvalue magnitudes correct; phases require calibration).

O-M27.2 (Conductor decoration formal import). The conductor $q_\chi \in \{1, 3, 11, 33\}$ is structurally separate from the $\mathfrak{so}(4)$ Cartan/chirality data per §4.3. Its operator-level realization within the Kostant framework requires importing the Connes (2000) conductor operator $\log|x|_v + \log|y|_v$ at finite places $v \in \{3, 11\}$, i.e., closure of ZS-M22 D4b OPEN. Status: OPEN; closure path = Connes–Burnol conductor operator import.

O-M27.3 (Sum rule Hodge interpretation, quantitative). The Hodge interpretation of the corpus sum rule $0.7948 + 0.2050 + 0.0001$ (§6.2) is HYPOTHESIS-strong. A quantitative match — i.e., explicit functional relationship between (harmonic / im D / complement) Hodge weights of the BV-BFV partition function and the FFPP scalar $|\lambda|^2 = (\pi^2/4) \cdot \eta_{\text{topo}}$ — would upgrade this to DERIVED. Status: OPEN; closure path = Cattaneo–Mnev–Reshetikhin partition function asymptotics on the explicit cobordism-history fiber.

O-M27.4 (Two V_4 groups: D_4 register vs Galois). ZS-F0 §8.6 PROVEN $\langle J, J_Z \rangle \cong D_4$ (register symmetry, acting on $\mathcal{H}_{\text{cob}}$) contains a $Z_2 \times Z_2$ subgroup that we identify as a V_4 . ZS-M22 §2.3 PROVEN $V_4 = \text{Gal}(K/\mathbb{Q})$ (arithmetic, acting on $\mathbb{C}[V_4]$) is structurally distinct. The Kostant Dirac framework couples them via D acting on $\mathcal{H}_{\text{cob}} \otimes \mathbb{C}[V_4]$, but explicit isomorphism between the two V_4 groups (e.g., as abstract groups or as subgroups of a larger ambient symmetry) is not established. Status: OPEN; closure path = Adams 2024 / Reduzzi–Xiao 2014 type arithmetic-geometric correspondence; or internal Z-Spin construction via BFV anchor structure.

§10. Non-Claims

This paper inherits non-claims NC-M22.X, NC-M23.1–7, NC-M24.X, NC-M25.1–6, NC-M26.1–7 verbatim. Six new non-claims are registered.

NC-M27.1: Does NOT claim a proof of the Riemann Hypothesis. NC-M23.1 preserved verbatim. The W3 closure of this paper is ONE of the THREE walls (W1, W2, W3) of the V_4 -equivariant ZBSI program; W1 and W2 remain OPEN under separate external imports (ZS-M26 §5.5).

NC-M27.2: Does NOT claim a proof of GRH for $L(s, \chi_{\{-3\}})$, $L(s, \chi_{\{-11\}})$, or $L(s, \chi_{\{33\}})$. The W3 closure provides the cohomological structure ($\dim H_D = 4 = |V_4|$ with one class per channel) required for the V_4 -equivariant Weil functional, but per-channel positivity remains tied to W2 (Pillar V) and external conductor decoration (D4b OPEN).

NC-M27.3: Does NOT claim closure of any Dragon D4 sub-target (D4a, D4b, D4c, D4d) of ZS-M23 §5.4. D4d (Cobordism BRST closure) is partially advanced — the BRST-Hodge harmonic projection $\Pi_{\text{Harm}} = \text{projection onto } \ker D$ is constructed — but the full D4d closure requires connecting H_D to the V_4 -decorated Sonin space $\mathcal{H}_{\text{Sonin}}^K$ of Connes–Consani–Moscovici 2024, which is D4a OPEN.

NC-M27.4: Does NOT claim that the conductor q_χ is internally determined by $\mathfrak{so}(4)$ representation theory. Per §4.3, q_χ is a separate arithmetic decoration encoding global Galois ramification of K , supplied externally via Connes–Burnol formalism.

NC-M27.5: Does NOT claim quantitative Hodge interpretation of the corpus sum rule $0.7948 + 0.2050 + 0.0001$. The §6.2 interpretation is HYPOTHESIS-strong; quantitative match registered as O-M27.3 OPEN.

NC-M27.6: Does NOT claim that $|\lambda|^2$ and the Kostant scalar $|\rho_{\text{gl}}|^2$ are connected by any closed-form identity. Per §6.3, they are independent structural inputs entering the construction at different axes (FFPP topological holonomy vs. Lie algebra root system invariant).

NC-M27.7: Does NOT introduce any new free parameter. All inputs are LOCKED from upstream corpus papers per Table 2.1, or IMPORTED from external peer-reviewed mathematics per §2.2. Anti-overclaim audit (test G1) PASSES.

§11. Conclusion

This paper closes Wall W3 of the V_4 -equivariant ZBSI program at **DERIVED-CONDITIONAL** status by importing the Kostant cubic Dirac framework. Three principal results were established:

(1) Theorem M27.1 (DERIVED-CONDITIONAL). On the cobordism-history fiber $V_{\text{Wilson}} \otimes \mathbb{C}[V_4]$ with $\mathfrak{so}(4) \cong \mathfrak{sl}(2)_L \times \mathfrak{sl}(2)_R$ structure, the Kostant cubic Dirac $D = \sum_a Z_a \otimes \gamma_a$ is the W3-closing BRST charge. $\dim H_D = 4 = |V_4|$ guarantees one cohomology class per V_4 channel (per Vogan-Huang-Pandžić 2002 [IMPORTED-2]). Conditional imports: ABM 2018 [IMPORTED-1]; Kostant 1999/2003 [IMPORTED-3].

(2) Theorem M27.2 (DERIVED). V_4 parity $a_\chi \in \{0, 1\}$ corresponds exactly to the Clifford chirality eigenvalue $\Gamma = \pm 1$ on the spinor module $S = \mathbb{C}^4$: even characters $\{1, \chi_{\{33\}}\} \leftrightarrow \Gamma = +1$, odd characters $\{\chi_{\{-3\}}, \chi_{\{-11\}}\} \leftrightarrow \Gamma = -1$. The conductor q_χ is registered as a separate arithmetic decoration (NC-M27.4).

(3) Theorem M27.3 (DERIVED-CONDITIONAL). The Cattaneo–Mnev–Reshetikhin modified quantum master equation $(\hbar^2 \Delta_{\text{BV}} + \Omega_{\text{BFV}}) \cdot \psi_\Sigma = 0$ is satisfied automatically for every $\psi_\Sigma \in H_D = \ker D$, with the chirality-graded decomposition $\Delta_{\text{BV}} \leftrightarrow D_+$, $\Omega_{\text{BFV}} \leftrightarrow D_-$. Conditional import: CMW 2018 [IMPORTED-4].

Two NEW Open Problems identified during the closure analysis are RESOLVED in this paper: O-M26.4 (non-trivial cohomology) and O-M26.5 (rank-1 vs rank-2 tension). The status of Corollary M26.3a is upgraded from HYPOTHESIS-strong to DERIVED-CONDITIONAL (under IMPORTED-1, IMPORTED-2, IMPORTED-3, IMPORTED-4).

Verification 24/24 PASS at machine precision. Five new falsification gates F-M27.1 through F-M27.5 registered; all PASS. Inherited ZS-M26 falsification gates F-M26.1 through F-M26.8 unchanged (all PASS). Cumulative falsification status for W3-closure direction: 13/13 PASS.

What this paper does NOT do. Per NC-M27.1, this paper does not prove RH or any GRH. The V_4 -equivariant ZBSI construction has THREE walls (W1 P3 closure under P1; W2 Pillar V V_4 Weil functional positivity; W3 cobordism BRST closure); only W3 is advanced here. W1 requires the Yakaboylu (2024) trace-norm convergence theorem (external, separate import). W2 requires the Connes (2000)–Burnol (2002, 2004) conductor/parity correction (external, separate import). The Z-Spin RH program continues to satisfy the principle of NC-M23.1: "Z-Spin does not, and structurally cannot, prove RH from internal data alone." What this paper adds is precise W3 closure under explicit external imports.

Code availability. Verification script `zs_m27_verify_v1_0.py` (Python 3.10+, NumPy, mpmath, SciPy) reproduces all numerical claims of this paper. Expected output: TOTAL 24/24 PASS, exit code 0.

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